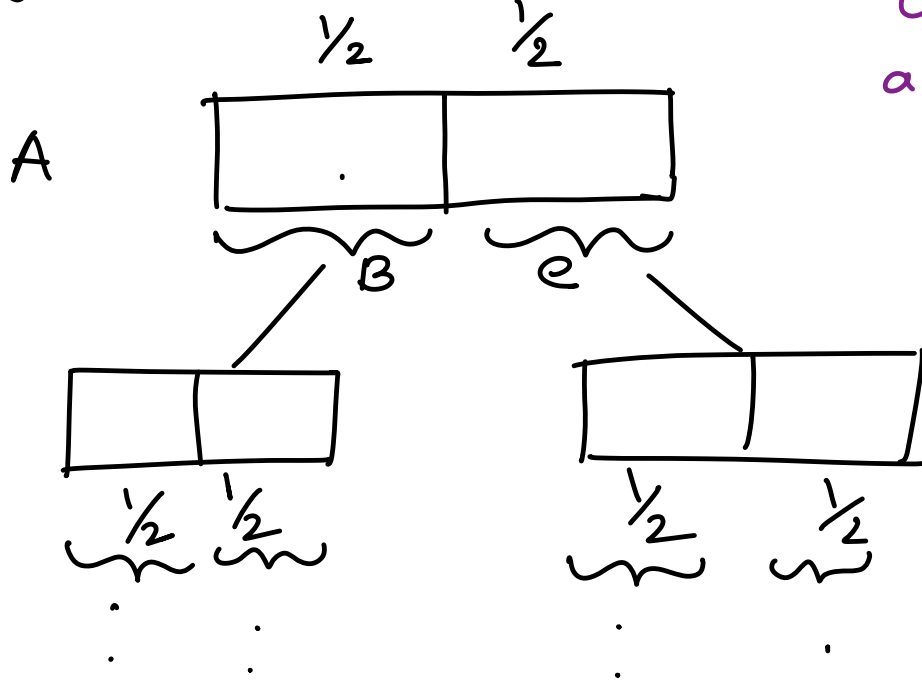
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Merge Sort

"Divide & Conquer" algorithm



Time Complexity = $\Theta(n \log n)$

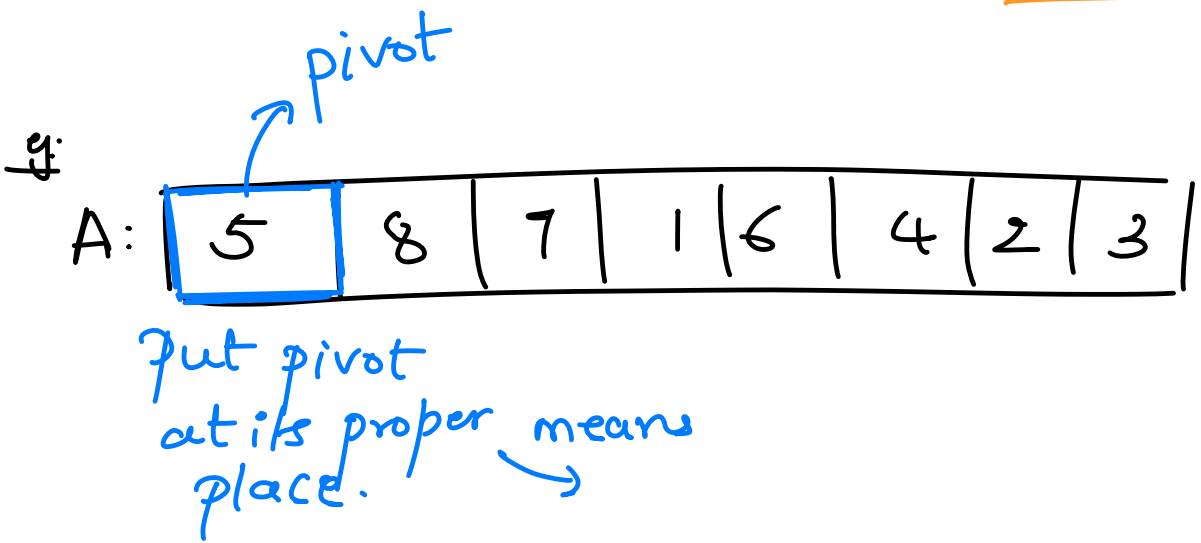
"Two auxiliary arrays are used (B & C).

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Q: Can you sort with $O(n \log n)$ time without using auxillary arrays.

Use divide & conquer
No auxillary arrays.

Yes.



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Step 1: Locate the 1st no $\leq x$ $A[i] = x$.

Exchange $A[2]$ with that number.

5	8	7	1	6	4	2	3
	$i=2$		$j=4$				

$A[i] \longleftrightarrow A[j]$

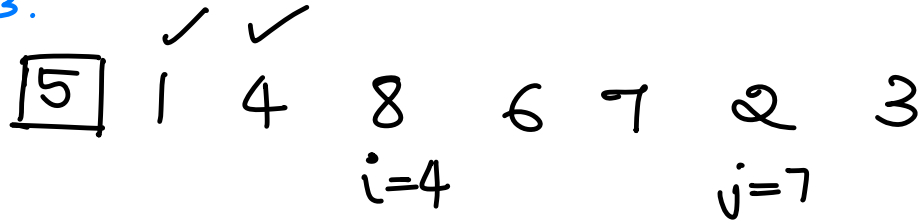
Step 2:

5	1 ✓	7	8	6	4	2	3
	$i=3$				$j=6$.		

(Incremented) $i \leftarrow i+1$

$A[i] \longleftrightarrow A[j]$

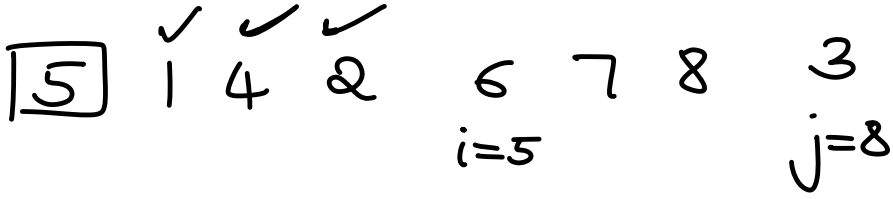
Step 3:



$i \leftarrow i+1$

$A[i] \leftrightarrow A[j]$

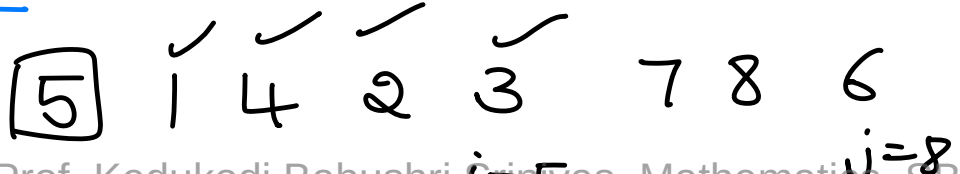
Step 4:



$i \leftarrow i+1$

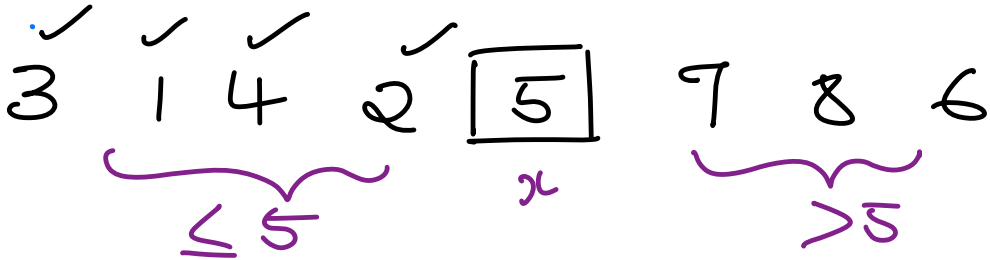
$A[i] \leftrightarrow A[j]$

Step 5:



$x = A[i] \longleftrightarrow A[j]$

(without incrementing i to i+1)



"Partitioning" Algorithm.

&

"Inplace" Algorithm.

(Inplace - no auxillary array used).

Partition (A, p, q) (1962)

1. $x \leftarrow A[p]$
2. $i \leftarrow p$
3. For $j \leftarrow p+1$ to q
4. If $A[j] \leq x$
5. $A[i] \leftrightarrow A[j]$?
6. $i \leftarrow i+1$
7. $A[p] \leftrightarrow A[i]$
8. return i

Note.

Merge Sort — "Merging is done repeatedly"

Quick Sort — "Partitioning is done repeatedly"

Quicksort (A, p, q)

1. If $p < q$
2. then $r \leftarrow$ Partition (A, p, q)
3. Quicksort $(A, p, r-1)$
4. Quicksort $(A, r+1, q)$

Initial Call: Quicksort $(A, 1, n)$.

Worst Case Analysis:

Worst Case Input : Descending order numbers (or ascending order).

eg: | 10 | 8 | 7 | 5 | 4 | 3 |

x

Quicksort (A, 1, n).

i p=1 2 3 4 5 6

eg: 3 4 5 7 8 10

x

j runs 2 to 6

No A[j] ≤ 3.

3 ⟷ 3.

p=2

4 5 7 8 10

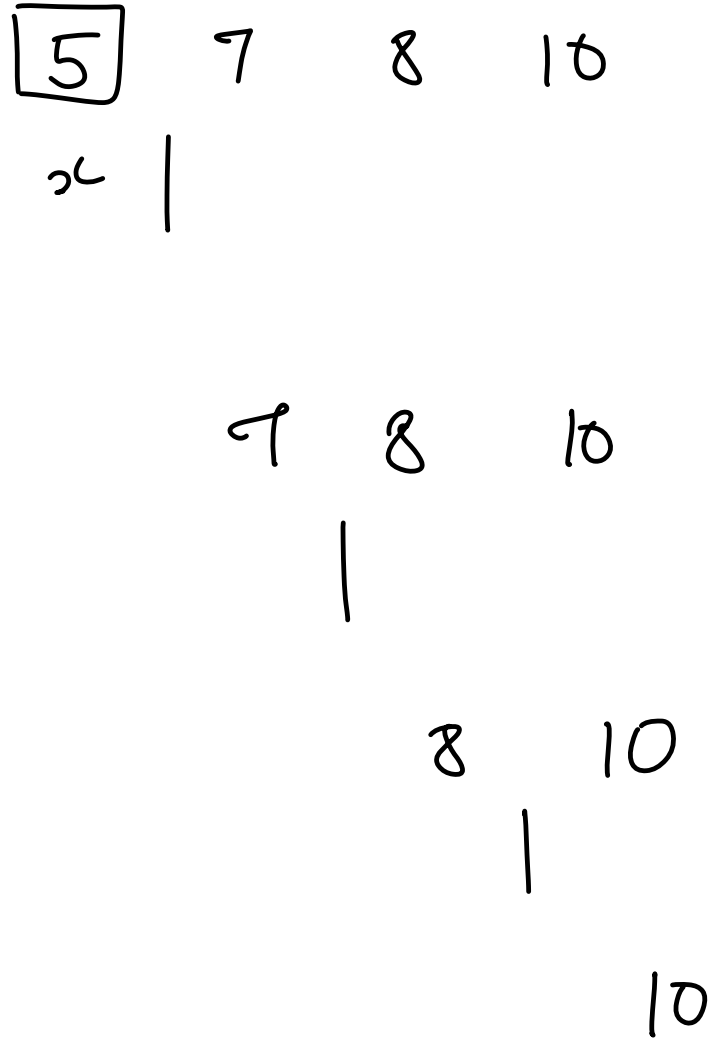
x

j runs from 3 to 5

No A[j] ≤ 4.

4 ⟷ 4

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5
7
8
10
7
8
10
8
10
10

Best Case:

Partitioning happening in the middle:

$$T(n) = \begin{cases} \Theta(1) & n = 1 \\ T(n/2) + T(n/2) + \Theta(n), & n > 1 \end{cases}$$

$\Theta(n)$: partitioning time.

Partition $T(n) = \sum_{i=1}^{n} 1 = n \in \Theta(n)$.

Worst Case Input:

Give already sorted array (in ascending order)

$$T(n) = \begin{cases} \Theta(1) & , \quad n = 1 \\ T(n-1) + T(1) + \Theta(n) & \text{if } n > 1 \end{cases}$$

Let $\Theta(n) = cn$, for some constant $c > 0$.

A:

$$T(n) = T(n-1) + T(1) + cn$$

$$= T(n-2) + T(1) + c(n-1) + T(1) + cn$$

$$= T(n-3) + T(1) + T(1) + T(1) + c(n-2) + c(n-1) + cn.$$

$$\vdots$$

$$= T(n-(n-1)) + \underbrace{T(1) + \cdots + T(1)}_{n-1 \text{ times}} + c(n-(n-2)) + \cdots + c(n-1) + cn$$

$$= T(1) + (n-1)T(1) + c(n + (n-1) + (n-2) + \cdots + 2 + 1)$$

$$= nT(1) + c\left(\frac{n(n-1)}{2}\right)$$

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$$= nT(1) + c\left(\frac{n(n-1)}{2}\right) \in \underline{\underline{\Theta(n^2)}}$$

Quicksort is quadratic time under worst case

Average Case Analysis: (Justification for why Quicksort is Quick)

Theoretical Average $\longrightarrow$ Expectation (using pdf)

$$\text{Average of } n \text{ numbers} = \frac{\sum_{i=1}^{n} x_i}{n}$$

$$= \frac{1}{n}x_1 + \frac{1}{n}x_2 + \cdots + \frac{1}{n}x_n .$$

pdf ?

Definition of pdf:

$p(x)$ is called a pdf if

$$1 \geq p(x_i) \geq 0 \quad \forall \quad i$$

$$\sum_{i=1}^{n} p(x_i) = 1$$

Average of n numbers $= \dfrac{\sum_{i=1}^{n} x_i}{n}$

$= \underset{\geq 0}{\frac{1}{n}} x_1 + \underset{\geq 0}{\frac{1}{n}} x_2 + \cdots + \underset{\geq 0}{\frac{1}{n}} x_n .$

$= E(X).$

$n > 0$

$1 \geq \frac{1}{n} > 0$ and $\sum_{i=1}^{n} \frac{1}{n} = 1$